

NADOR AL AROUI, Morocco

WMO: 603400

Lat: 34.983N

Lon: 3.017W

Elev: 581

StdP: 14.39

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.7	38.3	28.1	22.5	52.0	31.4	26.3	49.3	27.6	58.5	23.8	58.7	3.8	260	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.4	94.5	68.8	91.2	68.8	88.1	69.4	78.0	83.5	76.2	82.5	74.7	81.4	10.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	141.3	81.2	74.4	131.4	79.8	72.7	123.8	78.8	41.8	82.5	40.0	82.9	38.6	81.7	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	20.0	17.5	32.4	100.5	4.1	4.3	29.5	103.6	27.1	106.2	24.7	108.6	21.8	111.8
			31.0	79.1	4.3	4.2	27.9	82.0	25.4	84.5	23.0	86.8	19.9	89.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	53.1	54.6	57.4	60.7	65.7	72.1	77.3	78.5	73.3	67.7	59.8	54.8	(6)
		DBStd	9.70	4.73	4.73	4.13	3.61	3.85	3.88	3.71	3.59	3.85	4.32	5.25	4.52	(7)
		HDD50	47	22	12	2	0	0	0	0	0	0	0	1	10	(8)
		HDD65	1586	370	292	239	136	36	1	0	1	23	169	319		(9)
		CDD50	5386	117	141	230	321	486	663	845	883	698	549	295	158	(10)
		CDD65	1450	0	1	2	7	58	214	380	418	249	106	14	1	(11)
		CDH74	12267	2	7	35	95	449	1645	3590	4086	1726	543	84	5	(12)
		CDH80	4298	0	2	3	15	96	498	1457	1636	467	111	13	0	(13)
(14)	Wind	WSAvg	7.0	6.3	7.5	7.6	7.5	7.5	7.6	7.3	7.5	7.1	6.0	6.6	5.6	(14)
(15)	Precipitation	PrecAvg	13.5	1.5	1.4	1.5	1.3	0.8	0.2	0.1	0.2	0.8	1.5	2.0	2.0	(15)
		PrecMax	29.3	6.2	4.6	4.9	4.8	1.9	2.1	0.2	1.5	3.1	8.5	6.1	6.1	(16)
		PrecMin	7.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	(17)
		PrecStd	4.8	1.5	1.1	1.3	1.1	0.6	0.5	0.1	0.3	0.7	1.7	1.8	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	71.8	73.4	78.6	82.1	87.7	93.0	99.0	98.8	91.6	87.6	81.0	73.1		(19)
		0.4% MCWB	56.6	58.2	62.4	62.1	65.4	69.5	70.7	70.2	68.4	66.9	61.3	60.8		(20)
		2% DB	68.2	69.3	73.3	76.5	82.3	88.6	94.9	94.6	87.8	82.7	75.6	69.4		(21)
		2% MCWB	56.5	57.5	59.1	60.9	63.6	68.0	68.8	68.5	69.3	67.1	62.0	58.3		(22)
		5% DB	65.2	66.3	69.8	73.3	78.8	85.8	91.3	91.2	84.7	79.1	72.3	66.7		(23)
		5% MCWB	55.4	56.4	58.3	60.3	63.1	68.0	68.4	69.5	69.6	66.2	61.3	57.4		(24)
		10% DB	62.6	63.6	66.5	70.2	75.5	82.7	87.8	88.0	82.4	76.7	69.6	64.4		(25)
		10% MCWB	54.6	55.8	57.3	59.3	62.6	67.0	68.9	70.9	69.2	66.0	60.6	56.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	61.0	61.9	64.9	65.6	70.8	76.8	79.6	80.8	78.7	73.7	68.0	64.3		(27)
		0.4% MCDB	67.6	68.4	72.8	74.4	78.9	86.4	88.2	86.0	83.9	81.3	74.5	69.8		(28)
		2% WB	58.8	59.8	61.9	63.5	68.2	73.3	77.0	78.6	75.5	70.9	65.4	61.2		(29)
		2% MCDB	64.3	65.0	69.0	71.9	75.6	80.4	82.8	83.0	81.4	76.2	71.9	65.6		(30)
		5% WB	57.1	58.3	60.2	62.0	66.5	71.7	75.3	76.7	73.2	69.2	63.8	59.2		(31)
		5% MCDB	62.4	63.4	66.5	69.5	73.5	78.8	82.0	82.7	79.7	74.5	69.3	64.7		(32)
		10% WB	55.6	57.0	58.8	60.9	64.9	70.2	73.9	75.3	71.8	68.0	62.1	57.3		(33)
		10% MCDB	61.2	62.3	64.5	67.5	72.2	77.7	80.7	81.8	77.9	73.2	67.6	63.2		(34)
(35)	Mean Daily Temperature Range	MDBR	19.2	17.9	18.1	18.9	18.6	18.4	18.4	18.4	17.9	18.3	18.1	19.2		(35)
		5% DB MCDBR	23.3	22.5	25.9	25.4	26.0	24.8	26.3	25.8	23.4	23.5	22.3	21.8		(36)
		5% WB MCWBR	13.6	13.2	14.8	13.0	12.3	10.5	10.5	11.3	11.0	11.3	11.9	13.1		(37)
		5% WB MCWBR	19.1	17.7	20.1	19.8	18.4	17.6	15.8	15.5	16.3	16.6	17.1	18.1		(38)
(39)			12.4	11.7	13.1	12.1	10.8	9.9	8.8	8.8	9.7	9.5	10.6	12.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.346	0.403	0.414	0.447	0.481	0.521	0.505	0.460	0.410	0.356	0.324		(40)
		taud	2.493	2.406	2.225	2.215	2.149	2.053	1.943	2.001	2.137	2.271	2.417	2.493		(41)
		Ebn at Noon	283	284	274	277	268	258	246	248	253	257	263	272		(42)
		Edh at Noon	27	32	42	45	49	53	59	55	46	37	29	25		(43)
(44)	All-Sky Solar Radiation	RadAvg	908	1186	1555	1889	2168	2405	2357	2107	1710	1295	954	825		(44)
(45)		RadStd	74	99	124	133	160	72	56	74	73	83	74	59		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page